

NAVIOTECH SOLUTION PVT. LTD



AI INTERNSHIP PROJECT REPORT

ON

“BIG MART SALES PREDICTION USING MACHINE LEARNING”

Submitted in Partial Fulfillment of the Requirements of

Artificial Intelligence Internship Program

(Session: 2026-2027)

SUBMITTED BY

VINAY KUMAR OGRE

SUBMITTED TO

NAVIOTECH SOLUTION

NAVIOTECH SOLUTION PVT. LTD

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Naviotech Solution** for providing me with the opportunity to work on the project “**Big Mart Sales Prediction Using Machine Learning**” as part of the **Artificial Intelligence Internship Program**.

I am thankful to the mentors and guides for their continuous support, valuable suggestions, and encouragement throughout the project. Their guidance helped me understand the practical implementation of Machine Learning techniques and predictive analytics.

I would also like to thank my family and friends for their motivation and support during the successful completion of this project.

VINAY KUMAR OGRE

(Signature of the Candidate)

CERTIFICATE

This is to certify that **Mr. VINAY KUMAR OGRE** has successfully completed the project titled “**Big Mart Sales Prediction Using Machine Learning**” as part of the Artificial Intelligence Internship Program conducted by **Naviotech Solution**.

The project was carried out using Machine Learning techniques, including data preprocessing, feature engineering, model training, and sales prediction analysis using the **XGBoost** Regressor algorithm.

The work submitted is original and fulfills the requirements of the Artificial Intelligence Internship Program.

Date: _____

NAME

PROJECT GUIDE / MENTOR

NAVIOTECH SOLUTION

VINAY KUMAR OGRE

DECLARATION

I hereby declare that the project report entitled “Big Mart Sales Prediction Using Machine Learning” submitted as part of the Artificial Intelligence Internship Program is my original work.

The project has been completed by me under the guidance provided during the internship. The information presented in this report is true to the best of my knowledge and has not been submitted previously for any other certification or award.

Name: VINAY KUMAR OGRE

Date: _____

Signature: _____

ABSTRACT

The **Big Mart Sales Prediction** project is a Machine Learning based system developed to predict the sales of products sold in Big Mart stores. Sales prediction plays a significant role in business planning, inventory management, and decision-making processes. Accurate sales forecasting helps organizations optimize stock levels, reduce losses, and improve overall business performance.

This project utilizes historical sales data containing information about products and outlet characteristics. Various data preprocessing techniques were applied to clean and prepare the dataset. Missing values were handled, categorical variables were encoded, and relevant features were selected for model training.

The **XGBoost** Regressor algorithm was used to build the predictive model due to its efficiency and high prediction accuracy. The dataset was divided into training and testing sets, and the model was evaluated using the R^2 Score metric.

The obtained results demonstrate that Machine Learning can effectively predict product sales and support data-driven business decisions. This project highlights the practical application of Artificial Intelligence and predictive analytics in the retail sector.

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ARTIFICIAL INTELLIGENCE

INTRODUCTION

In today's competitive retail industry, predicting product sales is essential for effective business planning and inventory management. Retail companies need accurate sales forecasts to optimize stock levels, reduce wastage, and improve customer satisfaction.

Big Mart is a chain of retail stores that sells a wide variety of consumer products. Understanding the factors that influence product sales can help the company make better business decisions. Machine Learning techniques can **analyze** historical sales data and identify patterns that are difficult to detect manually.

The Big Mart Sales Prediction project aims to predict the sales of products across different Big Mart outlets using Machine Learning algorithms. The project utilizes historical sales data containing information about products, outlet characteristics, and sales performance.

Data preprocessing techniques such as missing value handling, data cleaning, and label encoding were applied to prepare the dataset. The **XGBoost** Regressor model was used to train the machine learning system and generate sales predictions.

This project demonstrates the practical application of **Artificial Intelligence** and **Machine Learning** in business analytics and predictive **modeling**.

EXISTING SYSTEM

In traditional retail businesses, sales forecasting is often performed manually using previous records and human judgment. Managers **analyze** past sales trends and estimate future demand based on experience.

Limitations of the Existing System:

1. Time Consuming
Manual sales analysis requires significant effort and time.
2. Inaccurate Forecasting
Human predictions may not accurately capture complex patterns in sales data.
3. Poor Inventory Management
Incorrect forecasts may result in overstocking or understocking of products.
4. Lack of Data-Driven Decisions
Traditional methods rely heavily on assumptions rather than data analysis.
5. Difficulty Handling Large Data
Manual analysis becomes challenging when dealing with thousands of products and transactions.
6. Reduced Business Efficiency
Inaccurate sales predictions can negatively impact profits and operational efficiency.

SYSTEM REQUIREMENTS

Software Requirements

- Operating System: Windows 10 / Windows 11 / Linux
- Programming Language: Python 3.x
- Development Environment: **Jupyter** Notebook / Google **Colab**
- Libraries Used:
 - NumPy
 - Pandas
 - Matplotlib
 - Seaborn
 - Scikit-learn
 - **XGBoost**

Hardware Requirements

- Processor: Intel i3 or higher
- RAM: Minimum 4 GB
- Storage: 500 MB free space
- Internet Connection (for Google **Colab**)
- Web Browser: Google Chrome / Microsoft Edge

OBJECTIVES OF THE PROJECT

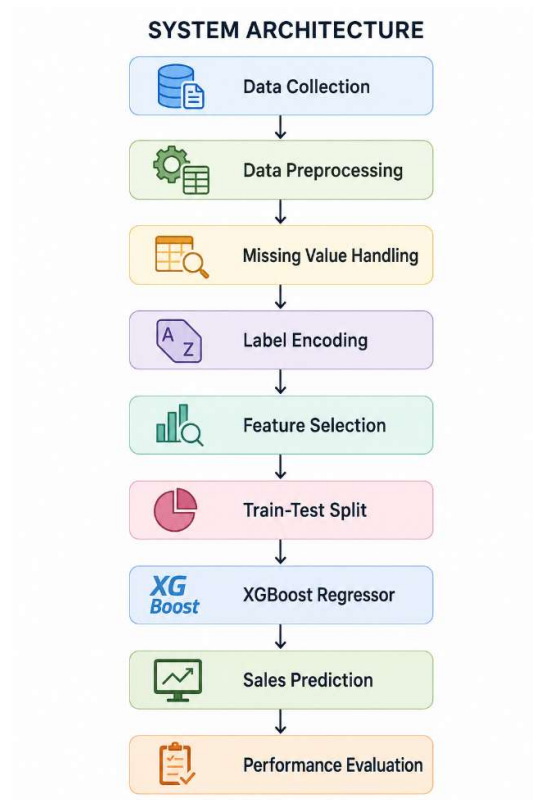
OBJECTIVES OF THE PROJECT

1. To predict product sales using Machine Learning techniques.
 2. To **analyze** historical sales data and identify important factors affecting sales.
 3. To perform data preprocessing and handle missing values effectively.
 4. To convert categorical variables into numerical form using label encoding.
 5. To train and evaluate a Machine Learning model using **XGBoost** Regressor.
 6. To improve forecasting accuracy and support business decision making.
 7. To understand the practical implementation of predictive analytics.
 8. To demonstrate the use of Artificial Intelligence in retail sales forecasting.
-

SYSTEM ARCHITECTURE

Architecture Description

- Data Collection: Dataset loaded from CSV files.
- Data Preprocessing: Cleaning and preparing data.
- Missing Value Handling: Filling missing values.
- Label Encoding: Converting categorical data.
- Train-Test Split: Dividing dataset into training and testing sets.
- **XGBoost** Model: Model training.
- Prediction: Sales prediction generation.
- Evaluation: R^2 score calculation.



SOFTWARE DESIGN

Data Preprocessing Design

- Missing value handling
- Data cleaning
- Feature transformation

Input Design

Input Features:

- Item Weight
- Item Fat Content
- Item Visibility
- Item Type
- Item MRP
- Outlet Size
- Outlet Type

Output Design

Output:

- Predicted Sales Value
- Training Score
- Testing Score

Model Design

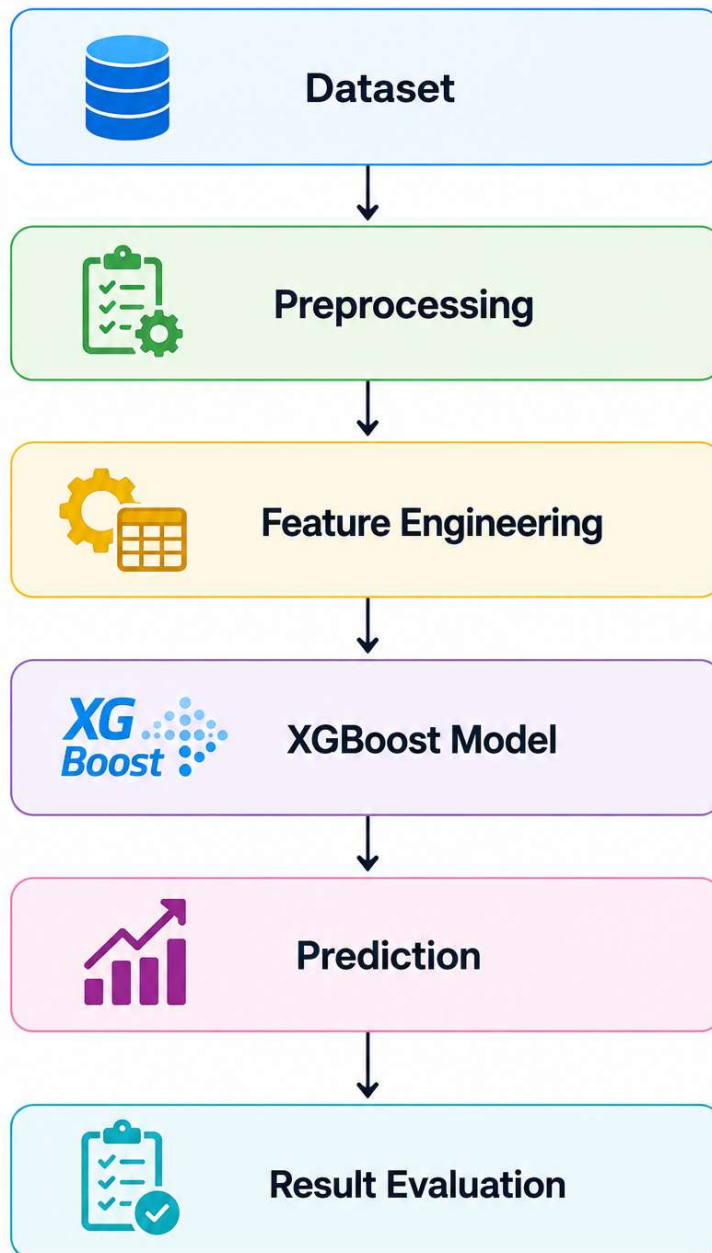
Machine Learning Algorithm:

XGBoost Regressor

Reason for Selection:

- High prediction accuracy
- Fast execution
- Handles large datasets efficiently
- Works well with structured data

SYSTEM FLOW DIAGRAM



DATASET DESCRIPTION

The dataset used in this project was provided as part of the Big Mart Sales Prediction task. The dataset contains historical sales information of products sold across various Big Mart outlets. It includes product-related attributes, outlet characteristics, and sales values.

The dataset is divided into two files:

1. Train.csv
2. Test.csv

The training dataset contains both input features and the target variable, while the testing dataset contains only input features used for prediction.

Dataset Features:

- Item Identifier – Unique identification number of the product.
- Item Weight – Weight of the product.
- Item Fat Content – Fat content category of the product.
- Item Visibility – Visibility percentage of the product in the store.
- Item Type – Category of the product.
- Item MRP – Maximum Retail Price of the product.
- Outlet Identifier – Unique identification number of the outlet.
- Outlet Establishment Year – Year in which the outlet was established.
- Outlet Size – Size category of the outlet.
- Outlet Location Type – Location category of the outlet.
- Outlet Type – Type of retail outlet.

Target Variable:

- Item Outlet Sales – Sales value of the product in a particular outlet.

The dataset contains thousands of records and provides sufficient information for training a Machine Learning model. Before model development, the dataset was cleaned and preprocessed to improve prediction accuracy.

PROPOSED SYSTEM

The proposed system is a Machine Learning based sales prediction system developed for forecasting product sales in Big Mart stores.

The system uses historical sales data and applies data preprocessing techniques to prepare the dataset. Missing values are handled, categorical features are encoded, and relevant attributes are selected for model training.

The **XGBoost** Regressor algorithm is used to build the predictive model. The trained model **analyzes** product and outlet characteristics and predicts future sales values.

Advantages of the Proposed System

- Improved prediction accuracy
- Data-driven decision making
- Better inventory planning
- Reduced forecasting errors
- Efficient handling of large datasets
- Faster business analysis

Working of Proposed System

1. Load Dataset
2. Perform Data Cleaning
3. Handle Missing Values
4. Apply Label Encoding
5. Split Dataset into Training and Testing Sets
6. Train **XGBoost** Model
7. Generate Predictions
8. Evaluate Performance using R^2 Score

IMPLEMENTATION

The implementation phase is one of the most important stages in any Machine Learning project. In this phase, the collected data is processed, **analyzed**, and used to build a predictive model capable of generating accurate results. The Big Mart Sales Prediction project was implemented using Python programming language and various Machine Learning libraries such as Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, and **XGBoost**.

The project began with loading the dataset into the Python environment. The dataset contained information about products, outlet characteristics, and corresponding sales values. Before building the machine learning model, it was necessary to understand the structure of the dataset and identify any data quality issues. Therefore, exploratory data analysis was performed to examine the dataset dimensions, column names, data types, and statistical information.

After **analyzing** the dataset, missing values were identified in certain columns such as Item Weight and Outlet Size. Missing values can negatively affect the performance of a machine learning model if not handled properly. To overcome this problem, appropriate preprocessing techniques were applied. Numerical missing values were replaced using mean values, while categorical missing values were replaced using the mode of the respective columns. This ensured that the dataset became complete and suitable for model training.

Once the missing values were handled, the next step was data transformation. The dataset contained several categorical features such as Item Fat Content, Item Type, Outlet Size, Outlet Location Type, and Outlet Type. Machine Learning algorithms cannot directly process text-based categorical data. Therefore, Label Encoding was applied to convert these categorical values into numerical form. This process transformed the dataset into a format that could be understood by the machine learning model.

Data visualization techniques were also used during implementation. Various graphs and distribution plots were generated using Matplotlib and Seaborn libraries. These visualizations helped in understanding the distribution of product characteristics and sales values. Visualization also helped identify patterns, trends, and potential relationships between different variables within the dataset.

After completing data preprocessing and transformation, the dataset was divided into independent variables and dependent variables. The input features included

product and outlet-related attributes, while the target variable was Item Outlet Sales, which represents the sales value that the model needed to predict. This separation allowed the machine learning algorithm to learn the relationship between input features and sales output.

The prepared dataset was then divided into training and testing datasets using the train-test split technique. The training dataset was used to teach the machine learning model, while the testing dataset was used to evaluate the performance of the trained model on unseen data. This approach helps measure how well the model generalizes to new data and prevents overfitting.

For the prediction task, the **XGBoost** Regressor algorithm was selected as the machine learning model. **XGBoost** is a powerful gradient boosting algorithm known for its high accuracy, efficiency, and ability to handle structured datasets effectively. The model was trained using the training dataset and learned the relationships between product attributes, outlet characteristics, and sales values.

Once the training process was completed, the model generated predictions for both training and testing datasets. To evaluate the performance of the model, the R^2 Score metric was used. The R^2 Score measures how well the predicted values match the actual values. A higher R^2 Score indicates better prediction accuracy and model performance.

The training score obtained from the model was approximately 0.8762, which indicates that the model learned the patterns present in the training dataset effectively. The testing score was approximately 0.5017, demonstrating that the model was able to make reasonably accurate predictions on unseen data. These results indicate that the machine learning model successfully captured important relationships within the dataset.

The implementation process demonstrated the complete machine learning workflow, including data collection, data preprocessing, feature transformation, model training, prediction generation, and performance evaluation. Each stage contributed significantly to the overall success of the project and ensured that the final model produced meaningful sales predictions.

This project provided practical experience in applying Artificial Intelligence and Machine Learning techniques to solve real-world business problems. It demonstrated how predictive analytics can help organizations forecast future sales, optimize inventory management, improve decision-making, and enhance business efficiency. The successful implementation of the Big Mart Sales Prediction system highlights the importance of data-driven approaches in modern retail and business analytics.

Data Collection and Processing

```
# loading the data from csv file to Pandas DataFrame
big_mart_data = pd.read_csv('/content/Train.csv')
```

```
# first 5 rows of the dataframe
big_mart_data.head()
```

1 to 5 of 5 entries Filter ?

index	Item_Identifier	Item_Weight	Item_Fat_Content	Item_Visibility	Item_Type	Item_MRP	Outlet_Identifier	Outlet_Establishment_Year	Outlet_Size	Outlet_Location_Type	Outlet_Type	Item_Outlet_Sales
0	FDA15	9.3	Low Fat	0.016047301	Dairy	249.8092	OUT049	1999	Medium	Tier 1	Supermarket Type1	3735.138
1	DRC01	5.92	Regular	0.019278216	Soft Drinks	48.2692	OUT018	2009	Medium	Tier 3	Supermarket Type2	443.4228
2	FDN15	17.5	Low Fat	0.016760075	Meat	141.618	OUT049	1999	Medium	Tier 1	Supermarket Type1	2097.27
3	FDX07	19.2	Regular	0.0	Fruits and Vegetables	182.095	OUT010	1998	Small	Tier 3	Grocery Store	732.38
4	NCD19	8.93	Low Fat	0.0	Household	53.8614	OUT013	1987	High	Tier 3	Supermarket Type1	994.7052

Show 25 per page

Like what you see? Visit the [data table notebook](#) to learn more about interactive tables.

Next steps: [Generate code with big_mart_data](#) [New interactive sheet](#)

```
[6]
✓ Os # checking for missing values
big_mart_data.isnull().sum()
```

```
...
0
Item_Identifier    0
Item_Weight      1463
Item_Fat_Content   0
Item_Visibility    0
Item_Type          0
Item_MRP           0
Outlet_Identifier   0
Outlet_Establishment_Year  0
Outlet_Size       2410
Outlet_Location_Type  0
Outlet_Type        0
Item_Outlet_Sales   0

dtype: int64
```

Handling Missing Values

Machine Learning Model Training

XGBoost Regressor

```
[38] regressor = XGBRegressor()
```

```
[39] regressor.fit(X_train, Y_train)
```

▼

```
XGBRegressor(base_score=None, booster=None, callbacks=None,
             colsample_bylevel=None, colsample_bynode=None,
             colsample_bytree=None, device=None, early_stopping_rounds=None,
             enable_categorical=False, eval_metric=None, feature_types=None,
             feature_weights=None, gamma=None, grow_policy=None,
             importance_type=None, interaction_constraints=None,
             learning_rate=None, max_bin=None, max_cat_threshold=None,
             max_cat_to_onehot=None, max_delta_step=None, max_depth=None,
             max_leaves=None, min_child_weight=None, missing=nan,
             monotone_constraints=None, multi_strategy=None, n_estimators=None,
             n_jobs=None, num_parallel_tree=None, ...)
```

Evaluation

```
[40] # prediction on training data
training_data_prediction = regressor.predict(X_train)
```

```
[41] # R squared Value
r2_train = metrics.r2_score(Y_train, training_data_prediction)
```

```
[42] print('R Squared value = ', r2_train)

R Squared value = 0.8762174618111388
```

```
[43] # prediction on test data
test_data_prediction = regressor.predict(X_test)
```

```
[44] # R squared Value
r2_test = metrics.r2_score(Y_test, test_data_prediction)
```

```
[45] print('R Squared value = ', r2_test)

R Squared value = 0.5017253991620692
```

OUTPUT

1. Dataset Head :

```
[2] # loading the data from csv file to Pandas Dataframe
big_mart_data = pd.read_csv('/content/train.csv')

[3] # first 5 rows of the dataframe
big_mart_data.head()
```

index	Item_Identifier	Item_Weight	Item_Fat_Content	Item_Visibility	Item_Type	Item_MRP	Outlet_Identifier	Outlet_Establishment_Year	Outlet_Size	Outlet_Location_Type	Outlet_Type	Item_Outlet_Sales
0	FDA15	9.3	Low Fat	0.016047301	Dairy	249.8092	OUT049	1999	Medium	Tier 1	Supermarket Type1	3735.138
1	DRC01	5.92	Regular	0.019278216	Soft Drinks	48.2692	OUT018	2009	Medium	Tier 3	Supermarket Type2	443.4228
2	FDN15	17.5	Low Fat	0.016760075	Meat	141.618	OUT049	1999	Medium	Tier 1	Supermarket Type1	2097.27
3	FDX07	19.2	Regular	0.0	Fruits and Vegetables	182.095	OUT010	1998	Small	Tier 3	Grocery Store	732.38
4	NCD19	8.93	Low Fat	0.0	Household	53.8614	OUT013	1987	High	Tier 3	Supermarket Type1	994.7052

Show 25 per page
Like what you see? Visit the [data.table.notebook](#) to learn more about interactive tables.

Next steps: [Generate code with big_mart_data](#) [New interactive sheet](#)

```
[4] # number of data points & number of features
big_mart_data.shape
```

2. Dataset Info

```
[5] # getting some information about the dataset
big_mart_data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8523 entries, 0 to 8522
Data columns (total 12 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   Item_Identifier        8523 non-null  object  
 1   Item_Weight            7060 non-null  float64  
 2   Item_Fat_Content       8523 non-null  object  
 3   Item_Visibility        8523 non-null  float64  
 4   Item_Type              8523 non-null  object  
 5   Item_MRP               8523 non-null  float64  
 6   Outlet_Identifier       8523 non-null  object  
 7   Outlet_Establishment_Year 8523 non-null  int64  
 8   Outlet_Size            6113 non-null  object  
 9   Outlet_Location_Type    8523 non-null  object  
10   Outlet_Type            8523 non-null  object  
11   Item_Outlet_Sales      8523 non-null  float64  
dtypes: float64(4), int64(1), object(7)
memory usage: 799.2+ KB
```

3. Missing Values

```
[9] # checking for missing values
big_mart_data.isnull().sum()
```

...	0
Item_Identifier	0
Item_Weight	1463
Item_Fat_Content	0
Item_Visibility	0
Item_Type	0
Item_MRP	0
Outlet_Identifier	0
Outlet_Establishment_Year	0
Outlet_Size	2410
Outlet_Location_Type	0
Outlet_Type	0
Item_Outlet_Sales	0

dtype: int64

Handling Missing Values

Mean --> average

Mode --> more repeated value

4. Data Cleaning

```
[15] # checking for missing values
big_mart_data.isnull().sum()
```

...	0
Item_Identifier	0
Item_Weight	0
Item_Fat_Content	0
Item_Visibility	0
Item_Type	0
Item_MRP	0
Outlet_Identifier	0
Outlet_Establishment_Year	0
Outlet_Size	0
Outlet_Location_Type	0
Outlet_Type	0
Item_Outlet_Sales	0

dtype: int64

Data Analysis

5. Label Encoding

```
Label Encoding

[30] ✓ Os encoder = LabelEncoder()

[31] ✓ Os big_mart_data['Item_Identifier'] = encoder.fit_transform(big_mart_data['Item_Identifier'])
      big_mart_data['Item_Fat_Content'] = encoder.fit_transform(big_mart_data['Item_Fat_Content'])
      big_mart_data['Item_Type'] = encoder.fit_transform(big_mart_data['Item_Type'])
      big_mart_data['Outlet_Identifier'] = encoder.fit_transform(big_mart_data['Outlet_Identifier'])
      big_mart_data['Outlet_Size'] = encoder.fit_transform(big_mart_data['Outlet_Size'])
      big_mart_data['Outlet_Location_Type'] = encoder.fit_transform(big_mart_data['Outlet_Location_Type'])
      big_mart_data['Outlet_Type'] = encoder.fit_transform(big_mart_data['Outlet_Type'])
```

6. Model Training

```
Machine Learning Model Training

XGBoost Regressor

[38] ✓ Os regressor = XGBRegressor()

[39] ✓ Os regressor.fit(X_train, Y_train)

XGBRegressor
XGBRegressor(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric=None, feature_types=None,
              feature_weights=None, gamma=None, grow_policy=None,
              importance_type=None, interaction_constraints=None,
              learning_rate=None, max_bin=None, max_cat_threshold=None,
              max_cat_to_onehot=None, max_delta_step=None, max_depth=None,
              max_leaves=None, min_child_weight=None, missing=nan,
              monotone_constraints=None, multi_strategy=None, n_estimators=None,
              n_jobs=None, num_parallel_tree=None, ...)
```

Evaluation

7. Final Result

```
[40]
✓ 0s      # prediction on training data
training_data_prediction = regressor.predict(X_train)

[41]
✓ 0s      # R squared Value
r2_train = metrics.r2_score(Y_train, training_data_prediction)

[42]
✓ 0s      print('R Squared value = ', r2_train)
✓
R Squared value = 0.8762174618111388

[43]
✓ 0s      # prediction on test data
test_data_prediction = regressor.predict(X_test)

[44]
✓ 0s      # R squared Value
r2_test = metrics.r2_score(Y_test, test_data_prediction)

[45]
✓ 0s      print('R Squared value = ', r2_test)
✓
R Squared value = 0.5017253991620692
```


TESTING

Model Training Testing

The **XGBoost** Regressor model was successfully trained using the training dataset.

- Data Validation Testing
- The dataset was checked for missing values, duplicate entries, and incorrect data types.
- Feature Encoding Testing
- Categorical variables were successfully converted into numerical values using Label Encoding.
- Prediction Testing
- The trained model generated sales predictions for unseen data.

Performance Evaluation

Training R^2 Score = **0.8762**

Testing R^2 Score = **0.5017**

- Results indicate that the model learned meaningful patterns from the dataset and achieved satisfactory predictive performance.

CONCLUSION

The Big Mart Sales Prediction project successfully demonstrates the application of Machine Learning techniques in retail business analytics. Historical sales data was **analyzed** and **preprocessed** using various data preparation techniques.

The **XGBoost** Regressor model was trained and evaluated using training and testing datasets. The model achieved satisfactory prediction performance and demonstrated the ability to forecast product sales based on product and outlet characteristics.

This project highlights the importance of data-driven decision making and the role of Artificial Intelligence in solving real-world business problems.

FUTURE WORK

1. Use larger datasets for improved accuracy.
2. Implement Deep Learning models.
3. Develop a web-based prediction dashboard.
4. Deploy the model on cloud platforms.
5. Integrate real-time sales data.
6. Create mobile applications for prediction.
7. Add advanced visualization features.
8. Improve forecasting accuracy using ensemble techniques.

BIBLIOGRAPHY / REFERENCES

1. Python Documentation – <https://docs.python.org>
2. Pandas Documentation – <https://pandas.pydata.org>
3. Scikit-Learn Documentation – <https://scikit-learn.org>
4. XGBoost Documentation – <https://xgboost.readthedocs.io>
5. NumPy Documentation – <https://numpy.org>
6. Seaborn Documentation – <https://seaborn.pydata.org>
7. Machine Learning by Tom M. Mitchell
8. Hands-On Machine Learning by Aurélien Géron